TECHNICAL SPECIFICATION FOR MOTORISED SLIDING HANAGR DOOR

1.0 GENERAL

The motorized bi-parting telescopic Multi leaf sliding Hangar Doors shall be designed, manufactured & erected from specialist firms.

The work includes design, fabrication, erection, testing and commission of Motorized Biparting sliding Hangar door. The motorized sliding hangar shall be designed to withstand the wind pressure and climatic conditions of Bhiwadi, Rajasthan.

DOOR CONFIGURATION

The sliding door shall be designed to provide minimum clear opening of 38.76m (W) x 7m (H). (See drawing - pdf file attached) The door shall be bi-parting with four number of leaves on each side, i.e., 8 leaves per entry. The hangar doors are needed at both front and rear of the hangar, i.e., 2 entries. When the door is operated for either OPEN or CLOSE positions each master leaf will drag remaining leaves and move to the parking bay or close position. Both the sections shall move to the extremities and automatically stop at its pre-set electrical limit switches.

A pass door / wicket door shall be provided of approximatesize 1000mm x 2000mmto enable personal access.

2.0 SCOPE OF WORK

Scope includes all labour, materials, equipment and services necessary to supply & install door based upon the drawings and the requirements of the specification.

3.0 DOOR DESIGN & CONSTRUCTION

3.1 WIND PRESSURE

The door leaves and tracks systems are designed to accommodate local wind speeds / pressures which are calculated in accordance with IS 875: 1987 (Part 3) "Wind Loads on Buildings & Structures". The basic wind speed to be used in the design consideration 25.0 metres per second or applied wind load of 1.5 kN/m2, whichever is more critical for design. In addition, further reference is made to ASCE 7-05 for guidance on internal pressure co-efficient.

3.2 STRUCTURAL DESIGN

Major structural elements of the sliding door system to be designed in accordance with IS 800: 2007 "Code of practice for general construction in steel".
Design shall be based upon the use of standard hot-rolled steel shapes - or equivalent "built-up" - (principally RHS/SHS/PFB/ UB/ IPF sections or smaller hollow sections).

The maximum deflection of any main structural member shall be limited to L/120 as measured over its whole length. Deflections of shorter segments of these main sections, and all other sections shall be much lesser than this to prevent excessive deflections of the framing around the over cladding panels.

3.3 DOOR LEAF CONSTRUCTION

Sliding door shall be fabricated with RHS 150x100 section as main members & middle members and braced with Round pipe of suitable dia, outside 40mm Thk. Puff Panel. Main structural members of the door leaf to be fabricated in accordance with EN 10210 Part 2: 2006 to minimum Grade S275 in accordance with EN 10025: 2004 and EN 10210 Part 1:1994 or equivalent Indian standards mentioned.

All parts shall be completely prefabricated before delivery to site so that all site connections are bolted to assemble the doors at site. All connections to be made using minimum Grade 8.8 bolts, zinc plated and de-embrittled, in accordance with ISO 898 Part 1.

Since the Hangar is air conditioned, 40 - 50 mm Puff Panels are to be provided on one side of the door leaf. The resulting door thickness will be the thickness of the Frame + thickness of Puff Panel.

3.4 BUILDING DEFLECTIONS

The sliding door system is to incorporate a top roller system such that it can accommodate maximum deflections of the roof structure immediately above the doors.

The track and top roller assembly of sliding door shall be coordinated such that the roller can also support itself whilst moving along the track and top tracks are able to accommodate live load deflections or deviations in the top tracks of up to L/360 (where "L" is the minimum roof span dimensions).

3.5 DOOR CLADDING

The cladding system covering the entire external face of the door leaves is to comprise the following basic elements:
Minimum 40mm thick. Chloro flouro carbon free self-extinguishing type Polyurethane Foam (PUF) panel of density 40 Kg/m^3 having both sides 0.5 mm thk GI precoated sheets.

Few parts of eachdoor-leafcladding shall be constructed with minimum 40mm thick polycarbonate panel to allow adequate amount of **natural light inside the hangar**. These panels shall stop 100% UV light penetration. The Panels shall have an insulation co-efficient value of 1.5 W/(m2K).The Panel shall be encased in suitable aluminum profiles.

3.6 PROTECTIVE COATING TO STRUCTURAL STEEL

All MS structural members of the sliding door shall be spray painted with two coats of Polyurethane paint over one coat of epoxy primer.
Protective Coating/Painting:
One coat of epoxy primer – minimum 25 microns.
Two coat of Polyurethane – minimum 60 microns.
Total Coating thickness - minimum 100 microns.

3.7 WEATHERSEAL SYSTEM

The weather sealing system for the sliding doors must be designed in conjunction with the cladding system to minimize sand ingress in particular, all horizontal seals shall be designed to accommodate all vertical deflections of the building and door head yet still maintain a proper seal.
In addition, all seal gaps are to be "finger-nip" safe, in accordance with the EN 12453: 2001 standard for power operated industrial doors.

The sealing system shall be designed so that no friction marks are left by the seals rubbing across the internal or external face of the main door cladding system.
All vertical joints between leaves on adjacent tracks and the end jambs are to be fitted with twin tapered EPDM wiper seals (color black) set in aluminum extrusions.
At the bottom, each leaf is to be fitted externally with a heavy-duty nylon brush seal, which is retained in aluminum extrusions.

3.8 HARDWARE – BOTTOM OF DOOR

Each door leaf is to be supported at the bottom on minimum of **2 Nos. Heavy duty double-flanged steel track wheels** of required diameter as per design requirement and wheel bearing Tapered rollers or spherical bearings, either internal or cartridge type, arranged so that both horizontal and vertical loads shall be transferred to the rail only through the bearing. Grease points are to be provided for wheel bearings.
Bottom rails will be made of **ISMC100 with M. S. Solid bar for Rail.**
The bottom rails shall be fixed on M.S Steel embedded in RCC and shall be capable of withstanding all type of dead and live load including transferred wind load for smooth transmission of movement through motorized mechanism.

Track cleaners shall be provided which is non-sparking device to clear light debris from rail head and wheel flange grooves as leaf is moved.

3.9 HARDWARE – TOP OF DOOR

The top track section itself shall be a **"UB"-section 175X85mm** which is hot-formed steel to minimum Grade S355 in accordance with EN 10025: 2004. The top track section shall be designed to accommodate all loads imposed upon it. Furthermore, local buckling of the section due to maximum roller loads on the walls of the section shall not be permitted.
Each panel supported at top with suitable dia. ball bearing top roller assembly **(Minimum 2 nos. per panel with rollers of suitable dia or 75mm dia)**
The track and top roller assembly are coordinated such that the roller can also support itself whilst moving along the track (against its own dead weight and wind loads.). roller shall be inherently corrosion resistant (i.e. Zinc plated Mild steel) and mounted on low friction/sealed bearings which will require no regular lubrication.

The top track system shall be designed to limit the inclination of the door leaves due to:
discontinuity of bottom rail;
high friction of bottom rail;
blockage of bottom rail

3.10 BUFFERS

Heavy duty energy absorbing buffer stops are to be fitted at the extremities of the bottom tracks to prevent leaves over-running the ends of their tracks during towing-mode operation. These are to be designed to eliminate any trip hazard that might otherwise occur.

4.0 AUTOMATION

4.1 DRIVE UNITS

3-Phase Electric Motor of suitable rating (Min. 3Kw & 10 Ton Capacity) shall drive each leaf or master leaf depending on site requirement with IEC classification **IP55** or above & minimum thermal protection of 155(F). The drive transmission between the motor and the drive wheel shall be through a chain-sprocket assembly.
Drive units should comprise a squirrel cage motor with integral magnetic brake and a high efficiency helical-bevel or helical-worm reducing gearbox, driving the bottom wheel(s) through a roller chain.
Each drive unit shall be fully enclosed in a removable casing which is easily accessible from the inside of the hangar via hinged or removable access covers. Drive unit access covers shall be interlocked so that the leaf cannot be driven under power unless all are closed.
The number and size of drive units shall be such that they are capable of safely operating their door leaf in still wind conditions whilst still meeting the following criteria:

They operate at 75% duty;

Drive units shall also be capable of operating the door safely at all wind speeds up to 20 metres per second (approximately 40 knots).

In the event that manual operation is required, a simple system which allows towing operation by powered vehicle e.g. fork truck, is to be provided

In case of power failure, the doors to be operated manually by disengaging the gearbox and manually operating the drive with the towing pin provided.

4.2 DOOR CONTROL SYSTEM

The sliding door control systems are to be designed to comply with EN 12453: 2001.
The variable frequency drive main electrical control equipment for each master leaf shall be housed in a standard steel electrical cabinet to IEC classification **IP65** or above as defined in EN 60529:1992. The cabinet shall be mounted inside the bottom section of the door leaf.
Each master leaf is to be fitted at each end with a pendant push button station. Push buttons and pendants shall be to IEC classification IP65 or above as defined in EN 60529:1992. The pendant is mounted to the end of the master door leaf on a cord about 2 metres long that allows the operator to move from side to side to see where the door is being driven. The pendants are to be hung up on hooks when not in use. Buttons provided should include (as a minimum):
"MOVE" button;
"EMERGENCY STOP" button;
"DOOR CLOSED" indicator light, to indicate when a leaf has reached its fully closed position.

Electronic devices should be provided which ensure that door leaves operating on the same track cannot collide with one another when being operated under power.

4.3 SAFETY AND WARNING DEVICES

Each leaf is to be fitted on the inside face with an **audible alarm and a visual alarm**. The alarms are to operate for a short period before a leaf starts to move.
Interlock switches are to be fitted to any access covers for drive units and electrical control panel doors, to prevent their leaf being used while these are open.
Touch safety edge that shall stop the door upon being pressed lightly and the door must immediately stop upon such contact.
Each leaf shall be provided with 4 nos of cleats (2 Top & 2 Bottom) designed of suitable dimension which shall be capable of withstanding the weight of the door including incase rollers become free/ fails /break then door shall not fall from its rails.

4.4 POWER SUPPLY AND WIRING MATERIALS

The incoming power supply which will be made available to the sliding doors is 400 volts 3 phase and earth, 50 Hz. The door Contractor is to advise the required current rating, taking account of voltage drops, start-up current surges etc.

4.5 AKAAP CONDUCTOR SYSTEM

Power is to be transmitted to each leaf by means of an enclosed duct type overhead conductor system. Each leaf shall draw a pick-up trolley through the conductor system with copper cable 4 core 4Sqmm, mounted in the top guide track area from a junction box placed at the top extremities inside the hangar. The conductor system shall be rated to IEC classification IP44 as defined in EN 60529:1992 and shall be capable of operating satisfactorily in maximum anticipated temperatures underneath the canopies.

All wiring within the leaves is to be Steel Wire Armour.

Prior to dispatch from the door manufacturer's Works, each drive unit and control panel is to be assembled into the bottom beam part of the doors, wired up as far as is possible and then test run in the Works.

5.0 DRAWING

Before fabrication of the motorized sliding hangar door, the general arrangements (GA) fabrication drawings & drawing of all sub-assemblies like control arrangements, interlocking / protection arrangements etc. shall be provided & approved by engineer in charge.

6.0 ADDITIONAL DESIGN FEATURES:

The following design features to be incorporated and elaborated in the proposal by Vendor

- Preventive method for hangar door not falling inside
- Alignment accuracy at the bottom rails; individually and collectively for the 4 rails; similarly, for the top rails
- Alignment accuracy between top and bottom rails
- Type of bearing at the bottom (to be able to take side loads better)
- Type of guidance at the top (specify schematics)

7. Scope of Vendor's Responsibility

The following Scope is for general guidance and indicative in nature, without prejudice to the technical specifications enumerated above. Specifics and deviations may be discussed and mutually agreed during execution:

i. Design, Manufacture, Fabrication, Assembly, Supply, Erection & Commissioning of Motorized Telescopic Bi-Parting Multi -leaf Sliding Hangar Door for ELCOM with Warranty/Guarantee- Guarantee of 36 months from date of Erection
ii. Complete Design, Analysis & Stress Reports
iii. Door leaf of RHS/SHS Sections - 8 leaves each x 2 = 16 leaves
iv. Structural polycarbonate panel for natural light shall be provided in each leaf as described in section 3.5 for cladding.
v. 40 - 50 mm puff panel (one side)- for 16 leaves
vi. Bottom Rail with Track, Top Track - for 16 leaves
vii. Bottom wheel (Suitable dia. 2Z SKF ball bearing): -2 set each leaf x 16 leaves.
viii. Top wheel (one wheel of 4 sub wheel, each of 75 mm dia): -2 set each leaf x 16 leaves
ix. Motorizations (Motor gear drive unit with chain sprocket assembly): - Four set for 16 panel for 2 Hangar enteries
x. Safety Hooter cum Flasher & buffers, gaskets, switches, VFD Control panel, Fire rated cables, EPDM Rubberseal to avoid contact from metal to metal, otherhardware: - complete for 2 hangar entries.
xi. Painting: -Two coats of Polyurethane paint over one coat of epoxy primer.
xii. Service door for man entry: - one each hangar entry
xiii. Buffers: - at end of each panel to work as shock absorber.
xiv. Track cleaner: - To keep dust / light weight items away from main track.
xv. Weather seals: - To protect from dust, rain.
xvi. Cleat: -To protect falling of door leaf in case of de railing.
xvii. Hooter cum flasher: - 70 dB(to work during operation of door).
xviii. Anti-crashing device - To prevent damage caused due to any obstacle while crossing the bottom track.
xix. Power feeding system: Enclosed duct type overhead conductor system

7. SCOPE OF RESPONSIBILITY OF HANGAR CIVIL CONTRACTOR

Vendor to clearly list out the items of responsibility of Hangar Civil Contractor for better co-ordination.